Used Car Resale Price Prediction Using Linear Regression

AI200 Project

Machine Learning Pipeline, Feature Engineering & Price Valuation Analysis

Course: AI200
Professor: SantoshKumar Vipparthi

Team Members:
Aashish (2025icb1433)
Yupendra Kumar (2025icb1452),
Yogendra (2025icb1451),
Bhavanam Harshavardhan Reddy (2025ICB1439),
Nikhil (2025icb1355)



Estimating resale value requires more than a single rule

Problem and objectives

Problem Statement: Estimating the resale value of a used car is complex because vehicle prices depend on multiple interacting factors, non-linear depreciation, wear and tear, and varying market values.

Core Objectives

- Analyze historical vehicle attributes and their impact on resale value.
- Perform statistical and Exploratory Data Analysis (EDA).
- Execute rigorous data cleaning, regex feature parsing, and leak-free preprocessing.
- Train a Log-Transformed Linear Regression model using Python and Scikit-learn.
- Evaluate model generalization using MAE, RMSE, R^2 score, and residual diagnostics.

Dataset overview and EDA

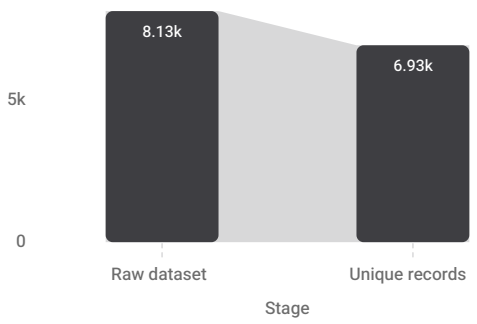
- **Source:** Kaggle — Vehicle Dataset from Cardekho (Car details v3.csv).
- **Raw structure:** 8,128 rows and 13 columns (7 categorical/string, 6 numerical).
- **Selling price:** ₹29,999 to ₹1,00,00,000; mean ₹6,38,271; median ₹4,50,000.
- **Other ranges:** year 1983–2020; km_driven 1–23,60,457 km; engine 624–3,604 CC; max_power 0–400 bhp; seats 2–14.
- **Categorical features:** name, fuel, seller_type, transmission, owner, mileage, and torque.
- **Visual insights:** selling price is heavily right-skewed; kilometers driven has a strong negative relationship with selling price.

Cleaning removed duplicates and corrected impossible values

85.21%

Total conversion rate

Records



Step by step

- **Duplicate removal:** 1,202 duplicate rows removed, reducing the dataset from 8,128 to 6,926 unique records.
- **Missing-data audit:** ~3% missing values across mileage (208), engine (208), max_power (206), torque (209), and seats (208).
- **Zero-value anomaly correction:** 15 records with 0.0 mileage and 3 records with 0.0 bhp max power converted to NaN for proper imputation.
- **Post-cleaning price distribution:** mean selling price ₹5,17,270; median ₹4,00,000.

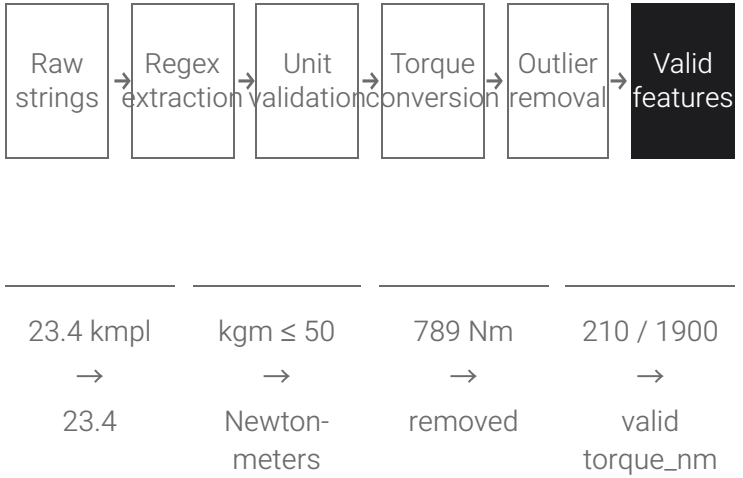
Regex parsing turned messy strings into usable model features

Mileage extraction and unit validation

- Extracted float values from strings, for example “23.4 kmpl” → 23.4.
- Cross-tabulation confirmed 100% unit consistency: all 6,616 Petrol/Diesel cars use kmpl, and all 87 CNG/LPG cars use km/kg.

Torque standardization

- Raw strings mixed Nm (6,227 rows) and kgm (481 rows) with RPM ranges.
- Regular expressions extracted numbers; kgm values ≤ 50 were converted to Newton-meters using 1 kgm = 9.80665 Nm.
- Impossible outliers above 700 Nm were removed, including 3 Maruti Zen D rows listed at 789 Nm.
- Edge-case recovery rules handled formats such as “210 / 1900” and “250@ 1250-5000rpm”, yielding 6,706 valid torque_nm records with mean 170.87 Nm.



Leak-free preprocessing expanded 10 inputs into 21 model features

- **Target:** selling_price.
- **Inputs:** year, km_driven, fuel, seller_type, transmission, owner, mileage, engine (CC), max_power (bhp), seats, and torque_nm.
- **Dropped:** name, raw torque, and mileage_unit.
- **Split:** 80% training (5,540 rows) and 20% testing (1,386 rows) from 6,926 cleaned records; random_state = 42.

- **Numerical pipeline:** median imputation.
- **Categorical pipeline:** most-frequent imputation plus OneHotEncoder(handle_unknown='ignore').
- The split happened before imputation, so test data never leaked into training statistics. Final processed matrix: 21 encoded features with 0 missing values.

The model explains 85.16% of unseen price variance

85.16%	₹1,06,764.98	₹1,80,389.30
R^2 score	MAE	RMSE

Model architecture and diagnostics

- **Log transformation:** LinearRegression trained on $y' = \ln(y)$; predictions mapped back with $\hat{y} = \exp(y')$ using TransformedTargetRegressor.
- **Fitted model:** intercept −202.87 across 21 feature coefficients; 0 negative or zero predictions; predicted test range ₹27,421 to ₹79,31,240.
- **Evaluation set:** 1,386 unseen test vehicles.
- **Residual statistics:** mean residual ₹23,290.61; minimum absolute error ₹16.09; maximum absolute error ₹28,12,818.56.

Error analysis and limitations

- Budget and mid-range vehicles priced between ₹1 Lakh and ₹15 Lakhs cluster tightly along the ideal diagonal.
- The 10 largest residuals occur in luxury and premium vehicles, including 2018–2019 Volvo, Lexus, BMW, and Mercedes-Benz models priced between ₹30 Lakhs and ₹55 Lakhs.
- **Key limitation:** dropping the name column avoided high-cardinality dimensionality, but the linear model could not explicitly capture luxury brand premiums beyond engine size, power, and torque specifications.

Team Roles & Contributions Summary

Member	Role	Contribution
Yogendra (2025icb1451)	Problem Selection & Direction	Defined problem statement, selected the Cardekho dataset, established project methodology, and guided analytical objectives.
Yupendra Kumar (2025icb1452)	Data Preprocessing	Removed 1,202 duplicates, audited missing/zero anomalies, extracted mileage units, and engineered the regex torque_nm standardization pipeline.
Aashish (2025icb1433)	Model Training & Testing	Executed the 80% training (5,540 rows) and 20% testing (1,386 rows) split, built the leak-free ColumnTransformer pipeline, and trained the TransformedTargetRegressor log-linear model.
Nikhil (2025icb1355)	Evaluation & Plots	Computed MAE, RMSE, and R^2 metrics (85.16%), conducted residual error analysis on luxury vehicle outliers, and created all EDA and diagnostic plots.
Bhavanam Harshavardhan Reddy (2025ICB1439)	Presentation & Synthesis	Structured project findings, synthesized technical workflows and metrics, and designed the final presentation deck.